

DESIGN AND DEMOCRACY

Design Thinking week6

LANGDON WINNER, THE WHALE AND THE REACTOR: A SEARCH FOR LIMITS IN THE AGE OF HIGH TECHNOLOGY 2020

Required Readings: 2nd Edition:

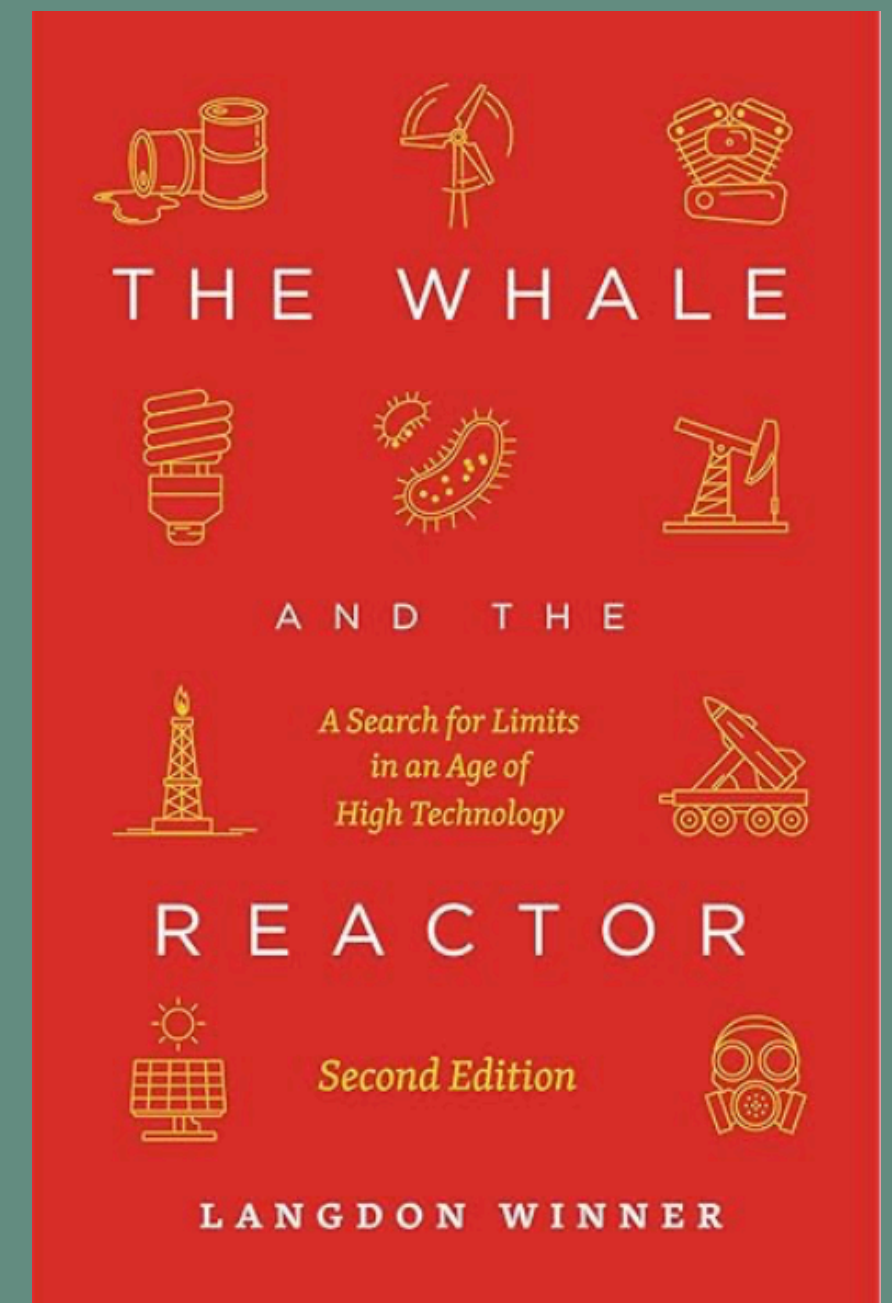
Chapter 1: Technologies as Forms of Life," pp.1-18

Chapter 2: Do Artifacts Have Politics? pp. 19-39; and

Chapter 3: Techne and Politeia," pp. 40-58.

Recommended Reading: 2nd Edition:

Chapter 11: Beyond Techno-Narcissism: Self and Other in the Digital Public Realm," pp. 179-196



DO ARTIFACTS HAVE POLITICS?

artifacts do have politics as they are not neutral tools but rather embody specific forms of power and authority, influencing human activity and social order

definition in term of Winner

Politics: distribution of power, authority, and privilege in a community

Technologies: a way of building order in our world -> anything that try to brings chaos in the world and try to bring order to it

2 MAIN WAYS THAT ARTIFACTS HAVE POLITICS

By decision

Technology it has the capacity to have political arrangement to it but it does not necessarily imply any particular political set up

Example: The bridges in long island New York

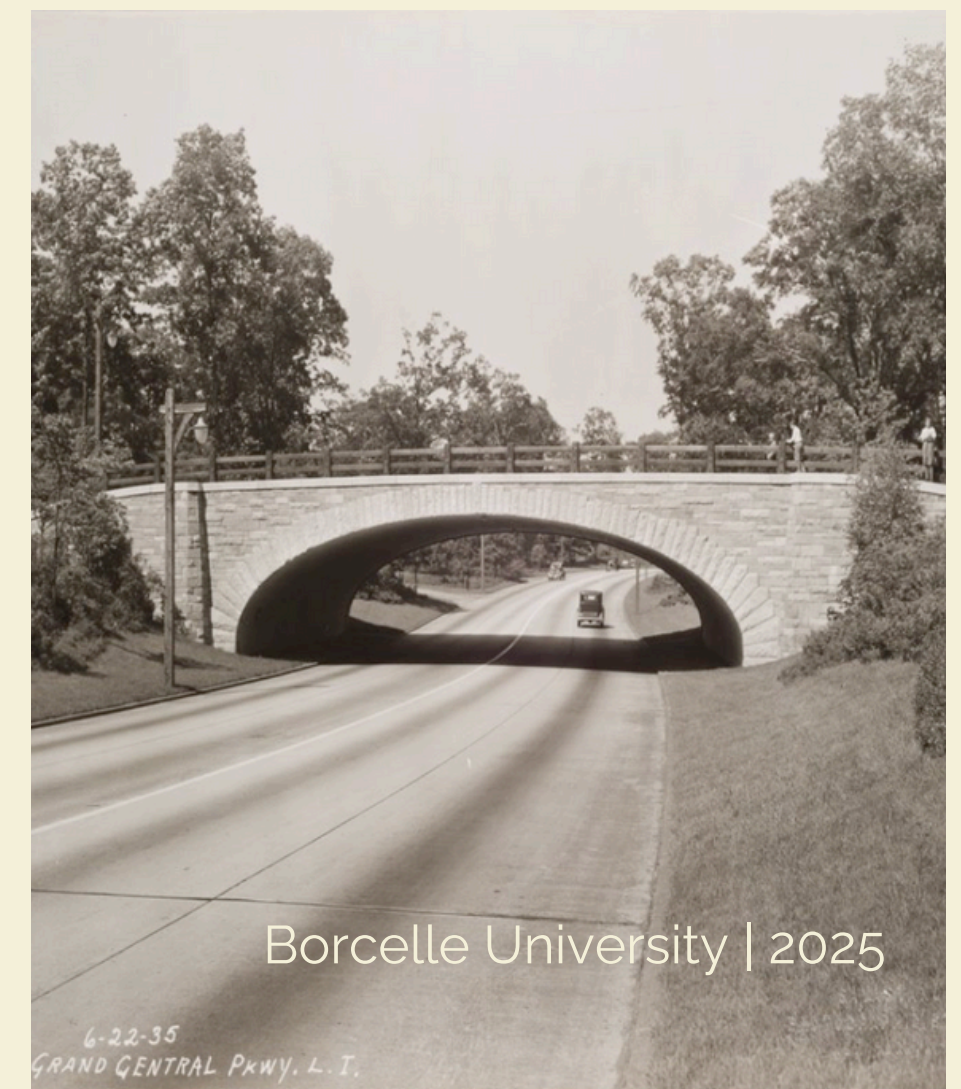
The designer created this bridges specifically out of class in racial bias
Wanted to keep poor and Black people from accessing certain areas.

Reasoning:

- Middle-class people = more likely to own cars (which could pass).
- Poor communities = more reliant on buses (which could not pass).

Key Point:

- The bridges didn't need to be that low.
- The design was a deliberate choice with social consequences.



Example: solar plant where much more conducive to a distributed manufacturing -> politics of technology come from human choices, not necessity. Solar energy can be built in many different ways, and its political effects depend on how both the hardware and supporting institutions are designed.

Flexible in design and arrangement

political meaning is determined by whether society chooses centralized power plants or locally controlled rooftop systems

By Necessity

Inherently political technologies

sometimes technology require a certain kind of political arrangement and maintenance of particular set of social conditions

Example: Nuclear energy require nuclear power plant -> require centralized authority -> need a huge area, specialize equipment, qualified people, certain infrastructure

Another example: atomic bomb, railroad, pipelines refineries -> big infrastructure things require/storing be compatible of necessity

Question before adoption

Should we even adopt it?

If yes, What form should it take?

When considering technology, Langdon Winner suggests that for **"inherently political technologies"** like nuclear power, the crucial question is "Should we even adopt it?" due to political consequences; the initial decision to adopt or reject them is decisive.

For technologies **"flexible in design and arrangement"** such as solar power, the primary concern is "What form should it take?" as different designs can lead to vastly different political consequences.

summary:

- Technical artifacts are not neutral; they carry built-in political qualities tied to power and authority.
- Winner distinguishes between two kinds of technologies:
- **Flexible technologies:** their design reflects social choices (e.g., Robert Moses's low bridges).
- **Inherently political technologies:** their very nature demands or aligns with specific power structures (e.g., atomic bombs, nuclear power plants).
- Adopting inherently political technologies is itself a decisive political decision

TECHNĒ AND POLITEIA

Defintions

Politeia: political order or proper political order of human relationships within a city-state

Techne: technology or something carefully designed

The reversal of a classic Analogy:

Tracing the historical analogy: Historically thinkers viewed political (politeia) as a specialized craft (techne) aimed at constructing a state

Plato and Rousseau compared politics to a craft, like shipbuilding where experts build a just society requiring skill and design to create a just constitution

Winner argues today this has flipped: modern technology itself now shapes society's structure, often more than politics does

Abundance and efficiency take over:

Historically, western worried luxury could destroy civic virtue (moral)

Overtime, the people came to believe that abundance driven by technological progress, inherently leads to freedom and happiness

- this perspective leads to neglect of specific form or design of technology which becomes unquestioned standard and underexamined criterion for progress
- allow more public issues and pushing aside democratic process

The unplanned technical constitution of society:

Over 200 years technology has silently created rules for society -> deciding power, authority, freedom, and justice without open debate

Key features:

- **centralize control:** tech like transport and communication gives huge power to governments, corporations, and the military
- **Gigantism:** large technologies need big organizations
- **Hierarchies:** tech systems often enforce rigid chains of command, reinforcing inequality
- **Eliminating alternatives:** small farming, craft work, and slower lifestyles get pushed out
- **Manipulation:** big organizations can manipulate external forces meant to control them such as control markets and public needs through practices like advertising

Lack of responsibility and call for limited:

The sociotechnical (inventors, entrepreneurs) have mostly prioritized profit over the impact on societal structure or justice

society rarely questions technology unless disasters are serious like “**cancer in laboratory animals**”

Example: The "dark side" of the Internet and social media, including computational propaganda, widespread misinformation "fake news"

Most discussion get reduced to vague “values” which hide real connections between technology and human well-being

A call for political Choice in Technology:

Winner argues we must consciously decide how technologies are designed and used, not just react to side effects **“responsibility for what we are making”**

we need democratic debate at the start of new technologies, rejecting those that go against freedom and justice

If not, we risk “sleepwalking” (technological somnambulism) into a future ruled by technology without real choice and democratic

summary:

- Technologies are not neutral tools; they actively shape the political and social order of society.
- Therefore, society must critically evaluate technologies beyond efficiency.
- Technologies should be judged by their compatibility with democratic values, including:
 - Freedom
 - Social justice
 - Self-government